Q1: 16 March (Shift 2) - Numerical

For real numbers α, β, γ and δ, if

$$\int \frac{(x^2-1)+\tan^{-1}\left(\frac{x^2+1}{x}\right)}{(x^4+3x^2+1)\tan^{-1}\left(\frac{x^2+1}{x}\right)}dx$$

$= \alpha \log_e\left(\tan^{-1}\left(\frac{x^2+1}{x}\right)\right)$
$+\beta \tan^{-1}\left(\frac{\gamma(\mathrm{x}^2-1)}{\mathrm{x}}\right)+\delta \tan^{-1}\left(\frac{\mathrm{x}^2+1}{\mathrm{x}}\right)+\mathrm{C}$

where C is an arbitrary constant, then the value

of $10(\alpha+\beta\gamma+\delta)$ is equal to ____

Q2: 18 March (Shift 1) - Single Correct

The integral $\int \frac{(2x-1)\cos\sqrt{(2x-1)^2+5}}{\sqrt{4x^2-4x+6}}dx$ is equal to

(where c is a constant of integration)

(1) $\frac{1}{2}\sin\sqrt{(2x-1)^2+5}+c$

(2) $\frac{1}{2}\cos\sqrt{(2x+1)^2+5}+c$

(3) $\frac{1}{2}\cos\sqrt{(2x-1)^2+5}+c$

(4) $\frac{1}{2}\sin\sqrt{(2x+1)^2+5}+c$

Q3: 18 March (Shift 1) - Numerical

If $f(x)=\int \frac{5x^8+7x^6}{(x^2+1+2x^7)^2}dx,(x \geq 0), f(0)=0$v and $f(1)=\frac{1}{\mathrm{K}}$, then the value of K is

Answer Key

Q1 (6) **Q2** (1) **Q3** (4)

Q1 (6)

$\int \frac{(x^2-1)dx}{(x^4+3x^2+1)\tan^{-1}\left(x+\frac{1}{x}\right)}+\int \frac{dx}{x^4+3x^2+1}$

$$\frac{\left(1-\frac{1}{x^2}\right)dx}{\left(\left(x+\frac{1}{x}\right)^2+1\right)\tan^{-1}\left(x+\frac{1}{x}\right)}+\frac{1}{2}\int \frac{\left(x^2+1\right)-\left(x^2-1\right)dx}{x^4+3x^2+1}$$

Put $\tan^{-1}\left(x+\frac{1}{x}\right)=t$

$$\frac{dt}{t}+\frac{1}{2}\int \frac{\left(1+\frac{1}{x^2}\right)\mathrm{dx}}{\left(x-\frac{1}{x}\right)^2+5}-\frac{1}{2}\int \frac{\left(1-\frac{1}{x^2}\right)dx}{\left(x+\frac{1}{x}\right)^2+1}$$

Put $x-\frac{1}{x}=y, x+\frac{1}{x}=z$

$\log_e t+\frac{1}{2}\int \frac{dy}{y^2+5}-\frac{1}{2}\int \frac{dz}{z^2+1}$

$$-\frac{1}{2}\tan^{-1}\left(\frac{x^2+1}{x}\right)+C$$

$$=\log_e \tan^{-1}\left(x+\frac{1}{x}\right)+\frac{1}{2\sqrt{5}}\tan^{-1}\left(\frac{x^2-1}{\sqrt{5}x}\right)$$

$\alpha=1, \beta=\frac{1}{2\sqrt{5}}, \gamma=\frac{1}{\sqrt{5}}, \delta=\frac{-1}{2}$

or

$\alpha=1, \beta=\frac{-1}{2\sqrt{5}}, \gamma=\frac{-1}{\sqrt{5}}, \delta=\frac{-1}{2}$

$10(\alpha+\beta\gamma+\delta)=10\left(1+\frac{1}{10}-\frac{1}{2}\right)=6$

Q2 (1)

$\int \frac{(2x-1)\cos\sqrt{(2x-1)^2+5}}{\sqrt{(2x-1)^2+5}}dx$

$(2x-1)^2+5=t^2$

$2(2x-1)2dx=2tdt$

$2\sqrt{t^2-5}dx = tdt$

So $\int \frac{\sqrt{t^2-5}\cos t}{2\sqrt{t^2-5}}dt = \frac{1}{2}\sin t + c$

$= \frac{1}{2}\sin\sqrt{(2x-1)^2+5} + c$

Q3 (4)

$f(x) = \int \frac{(5x^8+7x^6)dx}{x^{14}(x^{-5}+x^{-7}+2)^2}$

Let $x^{-5}+x^{-7}+2 = t$

$\left(-5x^{-6}-7x^{-8}\right)dx = dt$

$\Rightarrow f(x) = \int -\frac{dt}{t^2} = \frac{1}{t} + c$

$f(x) = \frac{x^7}{x^2+1+2x^7}$

$f(1) = \frac{1}{4}$